

Manzil: Intelligent Traffic & Emergency Routing

Project Proposal

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Abstract

Manzil is an AI-assisted traffic management and emergency routing system designed to improve the movement of vehicles through urban road networks. The system will analyze traffic conditions such as vehicle density, queue length and waiting time, and use this information to support adaptive traffic-signal decisions. A key feature is emergency response: when an ambulance is assigned to a destination, Manzil will evaluate available routes, select an optimal route according to current traffic conditions, and coordinate signals along that route to create a dynamic green corridor. The complete system will be developed and evaluated in a controlled SUMO based simulation with a web dashboard for visualization and analysis.

1. Introduction

Traffic congestion is a dynamic problem because traffic demand changes continuously across roads and intersections. Fixed-time traffic signals may give the same green duration to approaches even when their traffic demand is very different [2, 9]. Manzil addresses this problem by combining computer vision, traffic analysis and adaptive signal control in a simulated urban environment. Vehicle detection and tracking from traffic videos will provide traffic measurements using approaches such as YOLO and multi-object tracking [4, 5], while the simulation will allow the proposed control strategies to be tested safely and repeatedly [1, 2].

Emergency vehicles introduce an additional challenge. The shortest geographical route may not be the fastest route when roads are congested, or intersections have long signal delays. Manzil will therefore model the road network as a weighted graph and use dynamic travel costs to select an efficient ambulance route [2, 10]. As the ambulance moves, traffic conditions can trigger route recalculation. Signals on the selected path can then be coordinated to provide priority while maintaining safe signal transitions, following the principles of emergency vehicle signal pre-emption [10, 11].

2. Goals and Objectives

The main goal of Manzil is to develop and evaluate an intelligent traffic management prototype that can improve normal traffic flow and reduce emergency travel time. The project objectives are to:

- Develop a web dashboard and evaluate waiting time, queue length, throughput and ambulance travel time.
- Coordinate signals along the ambulance route to create a safe moving green corridor.
- Select an optimal ambulance route using dynamic traffic, queue and signal delay information, with rerouting when conditions change.
- Build a multi-intersection SUMO network and implement fixed time, adaptive and RL based signal control.
- Detect and classify vehicles from traffic video and estimate traffic density, queues and waiting time.

3. Scope of the Project

Manzil will focus on a manageable urban road network containing multiple connected intersections. A selected real world road area will be imported or reconstructed in SUMO so that the project can demonstrate realistic road connectivity while remaining computationally feasible [1, 3]. The system will generate normal, congested and emergency traffic scenarios and will compare different signal control strategies.

The traffic analysis component will use a YOLO-based vehicle detector with multi object tracking to obtain vehicle counts and movement information from recorded traffic video [4, 5]. The emergency routing component will accept origin and destination, calculate candidate routes using a weighted road graph, and select a route using estimated travel time, congestion, queue and signal-delay information [2]. If conditions change, the route can be recalculated. The signal controller will coordinate upcoming intersections to form a moving green corridor for the ambulance, consistent with emergency signal-priority approaches [10, 11].

A FastAPI backend will connect the traffic analysis, routing, control and simulation components, while a React-based dashboard will present the system state. SUMO and TraCI will be used to simulate vehicles and traffic lights and to allow the control algorithms to modify the simulation during execution [1, 2]. The project will remain simulation based and will not directly control real government traffic signals. Large scale city-wide deployment, live government CCTV integration and traffic forecasting are outside the current scope.

4. Initial Study and Work Done so Far

Initial study has focused on the feasibility and architecture of an AI-assisted traffic management system that can be demonstrated without access to real traffic-control infrastructure. SUMO was identified as the main simulation platform because it supports microscopic traffic simulation, traffic light control and external interaction through TraCI [1, 2]. OpenStreetMap based road network import provides a practical way to construct a representative urban network [3].

For traffic perception, public datasets such as BDD100K and UA-DETRAC provide annotated traffic scenes suitable for vehicle detection and tracking research [6, 7]. A pretrained YOLO model with tracking will be considered rather than training an object detector completely from scratch [4, 5]. For traffic control, the project will first implement fixed time and rule-based adaptive control and then investigate reinforcement learning using established traffic-signal control approaches and environments [8, 9, 12], so that improvements can be measured against clear baselines.

The emergency routing design combines graph-based route planning with dynamic traffic information. Literature on emergency vehicle route optimization, pre-emption and traffic signal coordination supports the idea of reducing intersection delay for emergency vehicles [10, 11]. At the current proposal stage, the team has finalized the problem definition, major features and proposed system architecture; implementation, dataset preparation and experimental evaluation will form the next development stages.

References

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